

13. Biodiversity and Conservation

- Biodiversity occurs not only in the species level, but also in the macromolecular levels.
- Biodiversity as described by Sociobiologist **Edward Wilson** is the combined diversity at all levels of biological organisation.
- The most important forms of biodiversity are:
 - Genetic diversity (diversity at the genetic level)
 - Species diversity (diversity at the species level)
 - Ecological diversity (diversity at the ecosystem level)

Genetic diversity (diversity at the genetic level)

- A single species might show high diversity at the genetic level in terms of the number of chromosomes (or genes) and variation in genes (allelic forms) even in different members of the same species, e.g.,
 - Number of genes in an *E. coli* - **4000**
 - In fruit fly, *Drosophila melanogaster* - **13,000**
 - The human, *Homo sapiens* - **30,000 – 40,000**
- The genetic variation (in term of allelic forms of the same gene) is expressed by the medicinal plant ***Rauwolfia vomitoria*** growing in different Himalaya ranges **in terms of the potency and concentration** of the active chemical (**reserpine** - is an indole alkaloid, antipsychotic and antihypertensive drug that has been used for the control of high blood pressure and for the relief of psychotic symptoms) that the plant produces.
- India has more than 50,000 genetically different strain of rice and 1000 varieties of mango.

Species diversity (diversity at the species level)

- Refers to the number and distribution of species in an area
- It is expressed in terms of '**number of species per unit area**' and also as '**number of individuals of different species in an area**'
- E.g., Western Ghats have a greater amphibian species diversity than the Eastern Ghats.

Ecological diversity (diversity at the ecosystem level)

- Refers to the diversity at ecosystem level. This can include variety of habitats, trophic levels, food webs, ecological processes and biotic interactions, etc
- Ecosystem diversity makes communities more productive and stable.
- E.g., India has greater ecosystem diversity (in terms of variety of habitats in deserts, rain forests, mangroves, coral reefs, wetlands, estuaries and alpine meadows) than a Scandinavian country like Norway
- **IUCN (International Union for Conservation of Nature 2004) - more than 1.5 million species have been recorded in the world**
- Out of total animals recorded, 70% (10,25,000 sps) are insects. It means that out of every 10 animals on this planet, 7 are insects
- About 6 million species on Earth waiting to be discovered and named
- The group fungi has more species than all the vertebrate species (fishes, amphibians, reptiles and mammals) combined
- Total number of species present on earth are considered to be about 7 million as estimated by **Robert May**.
- India has 2.4% of the world's land and 8.1% of the global species diversity. This is what makes our country one of the 12 mega diversity countries (India, China, Australia, Mexico, Columbia, Brazil, Madagascar, Indonesia, Malaysia, Peru, Ecuador, Venezuela) of the world.
- Nearly 45,000 species of plants and twice as many of animals have been recorded from India,

Patterns of Biodiversity

Latitudinal gradients

- The plants and animals are not distributed evenly worldwide. The diversity of living forms decreases as we go from the equator towards the poles. A huge amount of plants and animals are concentrated in the tropical region because of the following reasons.
- Tropical environment is less seasonal and almost constant and predictable as compared to temperate environment.
- Such constant environment promote niche specialization and lead to a greater species diversity.
- Tropics (latitudinal range of 23.5°N to 23.5°S) receive the major part of the solar energy, which contributes to great productivity.
- The largely tropical Amazonian rain forest in South America has the greatest biodiversity on the, Earth more than. 40,000 sps of plants, 3000 sps of fishes, 427 sps of amphibians, 378 sps of reptiles, 1300 sps of birds, 427 species of mammals, 125000 sps of invertebrates.
- Speciation is dependent upon time. Tropical areas have remained undisturbed for millions of years unlike temperate regions, which have experienced frequent glaciations in the past.

Species-Area relationships

- **Alexander von Humboldt** observed that biodiversity increases with increase in explored area.
- The relation between species richness and area, for a wide variety of taxa (angiosperm plants, birds, bats, freshwater fishes) is a **rectangular hyperbola**.
- On a logarithmic scale, the relationship is a straight line describe by the equation

$$\log S = \log C + Z \log A$$

Where,

S = Species richness

A = Area

Z = Slope of the line (regression coefficient)

C = Y - intercept

Value of Z is found to lie in the range of 0.1 to 0.2 for comparatively smaller areas such as countries while for very large areas such as entire continents, the slope of the line is much steeper with Z value lying from 0.6 to 1.2.

E.g., for frugivorous (fruit-eating) birds and mammals in the tropical forests of different continents, the slope is found to be 1.15.

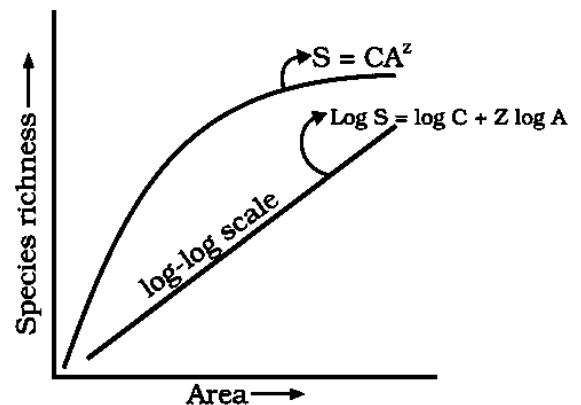


Figure: Showing species area relationship. Note that on log scale the relationship becomes linear

What is the importance of biodiversity on the Earth?

- Species diversity is important in term of the following.
 1. Stability
 2. Productivity
 3. Ecosystem health

1. Stability

- Communities with more species diversity, generally, trend to be more stable than those with less species. This is because such communities are more resistance or resilient to occasional disturbances (natural or man-made) and to invasions by alien species.

2. Productivity

- Ecosystems with higher species diversity are much more productive, E.g., Tropical forests.
- **David Tilman's** long-term ecosystem experiments showed that increased diversity contributed to higher productivity

3. Ecosystem health

- No species can exist independently. Interactions among species ensure good health and stability.
- In the long run, biodiversity is related with overall health of our ecosystem and survival of human race on the earth.

Importance of biodiversity

- Stanford ecologist **Paul Ehrlich** gave an analogy **rivet popper hypothesis** to help understand the importance of species. He compared each species with each rivet in the body of the airplane.
- This hypothesis explains that ecosystem to be an airplane and the species to be the rivets joining all parts together.
- If every passenger travelling in it starts popping a rivet to take home (causing a species to become extinct), it may not affect flight safety (proper functioning of the ecosystem) initially, but as more and more rivets are removed, the plane becomes dangerously weak over a period of time.
- Furthermore, which rivet is removed may also be critical.
- Loss of rivets on the wings (key species that drive major ecosystem functions) is obviously a more serious threat to flight safety than loss of a few rivets on the seats or windows inside the plane.

Loss of Biodiversity

- Due to human activities, the natural wealth is getting lost rapidly.
 - The **IUCN Red List / Red Data Book** (2004) documents the extinction of 784 species (including 338 vertebrates, 359 invertebrates and 87 plants) in the last 500 years
 - The last 20 years have seen the loss of 27 species.
 - Some of the causes of this loss are:
1. **Habitat loss and fragmentation** – this is the major cause for loss of biodiversity. Habitat destruction is caused by human activities such as deforestation and increasing pollution, leading to the loss of many plants and animals. Ex – Amazonian rain forest (**‘Lungs of the Planet’** – as they produce 20% of the world oxygen and provide the essential world environment service of continuously recycling CO₂ into O₂) earlier had 14% of the Earth’s land surface and reduced to less than 6%
 2. **Over-exploitation** – Humans due to their greed and increased exploitation of natural resources have contributed to the endangerment of commercially important species of plants and animals. Example – Species in the last 500 years such as **Steller’s sea cow and passenger pigeon** and many **marine fish** populations have **become extinct** due to over exploitation by humans. Some examples of recent extinctions include the Dodo (Mauritius), Quagga (Africa), Thylacine (Australia), Steller’s Sea Cow (Russia) and three subspecies of tiger – Bali, Caspian and Javan
 3. **Alien-species invasion** – the unintentional or deliberate introduction of alien species causes the declination of the indigenous species. Example – **Nile perch** introduced in **Lake Victoria** of South Africa led to the extinction of more than 200 species of **cichlid fish** in the lake. Extensive environmental damage caused and threat posed to our native species by invasive weed species like **carrot grass (Parthenium)**, **Lantana** and **water hyacinth (Eichhornia)**. The recent illegal introduction of the African catfish **Clarias gariepinus** for aquaculture purposes is posing a threat to the indigenous catfishes in our rivers.
 4. **Co-extinction** – when a plant or animal becomes extinct, another plant or animal which is dependent on it in an obligatory way also becomes extinct. Example – In case of plant pollinator mutualism, the extinction of one partner will eventually lead to the extinction of other also.

Biodiversity Conservation

- There are many reasons, some obvious and others not so obvious, but all equally important.
- They can be grouped into three categories:

1. **The narrow utilitarian**
2. **The Broadly utilitarian**
3. **Ethical**

1. **The narrow utilitarian** - Arguments for conserving biodiversity are that humans derive numerous direct economic benefits from
 - Food product (cereals, pulses and fruits)
 - Fire wood
 - Fibre (cotton, Jute from plants and silk wool from animals).
 - Construction material (timber for making furniture, houses, sports goods).
 - Industrial products (tannins, lubricants, dyes, resins, perfumes).
 - Products of medicinal importance (about 25,000 plants used in traditional medicine).
2. **The Broad Utilitarian** - We have a moral duty to care for their well-being and pass on our biological legacy in good order to future generations.
 - Pure oxygen – the Amazon forest is estimated to produce, through photosynthesis, 20% of the total oxygen in the Earth's atmosphere.
 - Pollination – without which plants cannot give us fruits or seeds, is another service, ecosystem provide through pollinators – butterflies, bees, bumblebees, birds and bats
 - Flood and soil erosion control occurs due to plants which help water retention, percolation and prevent soil erosion.
 - Nutrient replenishment is due to plant biomass that falls on ground is the biggest source of recycled nutrients after decomposition.
 - Waste recycling by microbes and other insects, etc.
 - Aesthetic pleasure that we derive from nature
3. **Ethical** - The ethical argument for conserving biodiversity relates to our moral obligation to conserve the planet that we share with millions of plant, animal and microbe species.
 - We need to realize that every species has an intrinsic value, even if it may not be current or any economic value to us.
 - We have a moral duty to care for their well-being and pass on our biological legacy in good order to future generations.

How do we conserve biodiversity?

- Conservation of biodiversity means protection, judicious and minimal use and rebuilding the damage units.
- There are two basic approaches in the conservation of biodiversity. These are.
 1. **In situ conservation**
 2. **Ex situ conservation**

1. In-situ conservation or On site conservation

- Involves protection of species in their natural habitat. This type of conservation includes following types

Hot spots – are the regions of high levels of species richness and high degree of **endemism** (that is, species confined to that region and not found anywhere else). These hotspots are also regions of accelerated habitat loss.

- Endemic species are confined only to a limited region.
- There are **34 hotspots** in the world
- **Three** of these **Hotspots in India** are given below which cover our country's exceptionally high biodiversity

- **Western Ghats and Sri Lanka**
- **Indo-Myanmar**
- **Eastern Himalaya**

- Although all biodiversity hotspots put together cover less than 2% of the earth's land area, the number of species they collectively harbour is extremely high and strict protection of these hotspots could reduce the ongoing mass extinctions by almost 30%.

Protected areas – are ecologically unique and biodiversity-rich regions that are legally protected as biosphere reserves, national parks and wildlife sanctuaries.

- India has **14 (18) biosphere-reserves, 90 (103) national parks** and **448 (515) wildlife sanctuaries**.
- **Sacred groves** - There were areas in forests that were set aside and all the trees and wildlife within were venerated and given total protection. Such sacred groves are found in
 - Khasi and Jaintia Hills in Meghalaya
 - Western Ghat regions of Karnataka, Maharashtra
 - The Sarguja, Chanda and Bastar areas of Madhya Pradesh
 - Aravalli hills of Rajasthan

2. Ex-situ conservation or Offsite conservation

- The threatened species of plants and animals are taken out of their habitats and are kept in special settings such as in zoological parks, botanical gardens, and wildlife parks.
- **Cryopreservation of gametes** – of threatened species and keeping them preserved in viable and fertile condition for long periods at very low temperatures
- **In vitro fertilisation** – to propagate endangered species. Offsprings can be produced by using preserved sperm to fertilize the eggs in vitro and then implanting in female animals
- **Tissue culture** – can be used to propagate plants species
- **Seed banks** – seeds of different genetic strains of commercially important plants can be kept for long periods, under specific conditions.
- **Gene banks** – is a comprehensive, public database that stores vast amount of DNA and RNA sequences alongside their protein translation.

Convention on Biodiversity

- The historic convention on biological diversity (the earth summit) held in **Rio De Janeiro** in 1992, called upon all nations to take appropriate measures for conservation of biodiversity and sustainable utilization of its benefits.
- The World Summit on Sustainable development held in 2002 in **Johannesburg, South Africa**, 190 countries pledged their commitment to achieve by 2010, a significant reduction in the current rate of biodiversity loss at global, regional and local levels.